

ASSIGNMENT 3

INDIA'S CONTRIBUTION TO TRIGONOMETRY

- Systematic study of trigonometric function began in "Hellenistic mathematics" reaching India as part of Hellenistic astronomy. In Indian astronomy, the study of trigonometric functions flourished in the Gupta period, especially due to Aryabhata (sixth century) (E), who discovered the 'sine' function.
- The word 'trigonometry' was derived from Greek $\tau\rho\iota\gamma\omega\upsilon\nu$ 'trigōnon', "triangle" and $\mu\epsilon\tau\rho\upsilon$ 'metron', "measure". The modern words "sine" and "cosine" are derived from the Latin word sinus. The word tangent comes from Latin tangere meaning "touching", since the line touches the circle of unit radius, whereas 'secant' stems from Latin 'secans' "cutting" since the line cuts the circle.
- The words Jya and Kojya became sine and cosine respectively.

$$\text{ज्या (अ)} = \text{sine (A)} = \sin(A)$$

$$\text{कोज्या (अ)} = \text{cosine (A)} = \cos(A)$$

$$\text{स्पृश (अ)} = \text{tangent (A)} = \tan(A)$$

In the 7th century, Bhaskara I produced a formula for $\sin(x)$ which had a relative error of less than 1.9%.

$$\sin x \doteq \frac{16x(\pi - x)}{5\pi^2 - 4x(\pi - x)}, \quad 0 \leq x \leq \pi$$

Later in 7th century, Brahmagupta redeveloped the formula

$$1 - \sin^2 x = \cos^2 x = \sin^2\left(\frac{\pi}{2} - x\right) \text{ and the}$$

Brahmagupta interpolation formula for computing sine values.

Another later Indian author on trigonometry was Bhaskara II in 12th century. Bhaskara II developed 'spherical trigonometry' and ~~dis~~ discovered many trigonometric results. Bhaskara II was one of the first to discover $\sin(a+b)$ and $\sin(a-b)$.
trigonometric results like:
 $\sin(a-b) = \sin a \cos b - \cos a \sin b$

Madhava (c. 1400) made early strides in the analysis of trigonometric functions and their infinite series expansions. He developed the concepts of power series and Taylor series and produced the power series

expansions of sine, cosine, tangent and arc-tangent. Using the Taylor series approximations of sine and cosine, he produced a sine table to 12 decimal places of accuracy and a cosine table to 9 decimal places of accuracy. He also gave the power series of π and the angle, radius, diameter, and circumference of a circle in terms of trigonometric functions. His works were expanded by his followers at the 'Kerala school' upto the 16th century.

NO.	SERIES	NAME
1.	$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	Madhav sine series Newton (1670)
2.	$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	Madhav cosine series Newton (1670) Leibnitz (1676)
3.	$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$	Madhav's arctangent series James Gregory (1671) and Leibnitz (1676)

→ The Indian text, the *Yukti bhāṣā* contains proof for expansion of the sine and cosine functions and the derivation and proof of the power series for inverse tangent, discovered by Madhava. The *Yukti bhāṣā* also contains rules for finding the sines and the cosines of the sum and difference of two angles.

→ Note: Aryabhata is the father of Indian Mathematics who significantly contributed to mathematics and astronomy and discovered theories like solar system rotations, approximation of trigonometric functions, place value system and zero and many others.

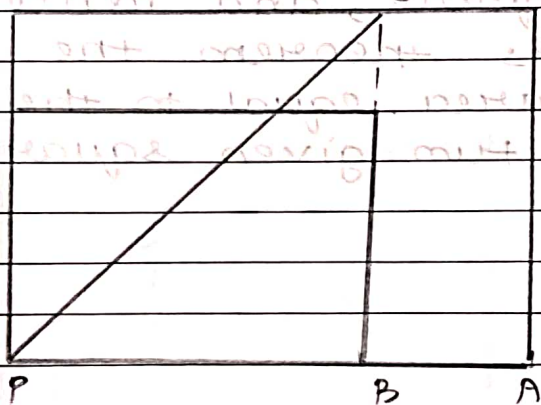
* APPLICATION OF PYTHAGORAS THEOREM

- To describe a square whose area is the sum of areas of two given squares.

- This is converted in two parts, one when the given squares are of equal size, another when they are unequal.

⇒ SQUARE WITH AREA EQUAL TO THE SUM OF TWO GIVEN SQUARES

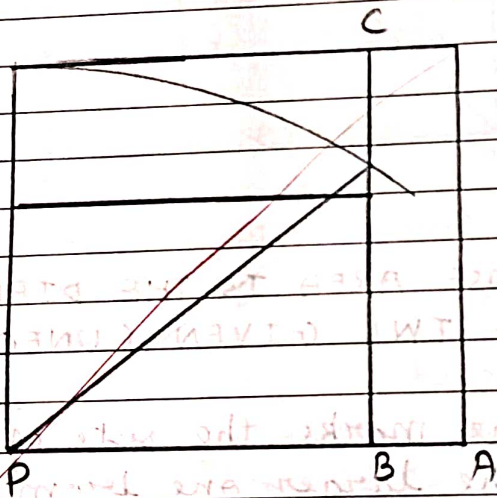
- The prescription is to mark the side of the smaller square on one of the sides of the larger square (from one of the vertices) and take the diagonal of the rectangle formed by it with the other side of bigger square.
- The square on this diagonal has area equal to the sum of the 2 given squares.



⇒ SQUARE WHOSE AREA IS THE DIFFERENCE OF AREAS OF TWO GIVEN (UNEQUAL) SQUARES.

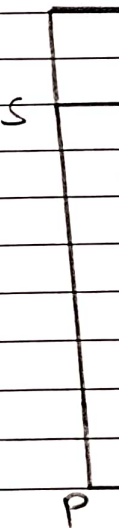
- For this one marks the side of the smaller square in the larger one from a corner, forming a rectangle as before.
- An arc is drawn with one of the vertices of the rectangle as the centre and the longer side as the radius, meeting the opposite side of the rectangle.

The portion of that side that is cut out from the vertex, nearer the point is picked, named as centre is the answer. BC is perpendicular for the side of the difference of the squares with sides PA and PB . We observe that the cut out portion together with the smaller side of the rectangle are sides of a rectangle whose diagonal equals the side of the bigger square, and hence by the Pythagoras theorem the square over it has area equal to the difference of the two given squares.



TO TRANSFORM
SQUARE W.

For this put
the smaller
rectangle
parts parallel
farther to
adjacent
square.



ut out
t is picked
BC is the
ference
d PB.

* TO TRANSFORM A RECTANGLE INTO A
SQUARE WITH THE SAME AREA.

For this purpose, forming the square over
the smaller side, the remaining
rectangle is divided into two equal
parts parallel to the square and the
farther part is taken off and put
adjacent to the other side of the
square.

